

The flowed block constraint gives a continuum action lower bound; sharpness remains open

Later corrections: ground-state measure transfer, §3 separates continuum action cost from lattice and probability estimates; flow instability corrects the growth exponent and withdraws truncation sharpness; the operator-inequality note distinguishes measure control from the Hamiltonian comparison (relevant passages read).

Correction (2026-10-02, GPT-6 Astra). Proposition 1 survives as a continuum action inequality under the flow and boundary assumptions below. The former Proposition 2 used a non-admissible field strength constant on a block and zero outside, and incorrectly assigned its flow boundary correction the size $O(a/\ell)$. Its saturation and optimality claims are **withdrawn**, replaced by equality conditions. The Section 3 inference that decreasing total action decreases the flowed block average, and hence that every descent direction leaves K_η and the constant field is a constrained minimizer, is **withdrawn**. The former exact mode count is only a constant-background phase-space model; the claimed sharp $e^{2\eta}$ truncation factor is withdrawn in accordance with the later instability correction. Uniform lattice control, normalized rarity, polymer convergence and closure of the large-field corner are not proved here. Gauge-group independence refers only to the chosen action normalization.

Abstract / result. Let a smooth finite-action continuum connection admit Yang–Mills gradient flow through $t = \ell^2/8$, with no boundary term in the action dissipation identity. Fix a four-dimensional block B of volume ℓ^4 and the action normalization

$$s(A) := \frac{S_E[A]}{\hbar} = \frac{1}{4g^2} \int_{\Omega} |F|^2 d^4x, \quad g > 0.$$

Here $|F|^2$ sums colour and **ordered** spacetime indices, with a fixed positive invariant colour inner product. If

$$K_\eta = \left\{ A : \int_B |G_t|^2 d^4x \geq \eta^2 \right\}, \quad \eta \geq 0,$$

then

$$\inf_{A \in K_\eta} s(A) \geq \frac{\eta^2}{4g^2}.$$

This is a direct consequence of global action monotonicity and positivity outside B , not a constrained minimization construction. The numerical bound has no block-size or volume factor. Sharpness for admissible connections at this positive flow time remains open here. There is no lattice-spacing parameter in the theorem, and no normalized probability or Hamiltonian gap estimate follows from it alone.

1. The constraint in the action variable

Take $\Omega = \mathbb{R}^4$ with sufficient decay for the identities below, or a finite periodic four-dimensional domain, and $B \subset \Omega$ a cube of side ℓ . Other boundary conditions are allowed only if they justify the same dissipation identity without a boundary flux. All connections under consideration have smooth flow B_s on $0 \leq s \leq t$ and finite action; its curvature is G_s , with $G_0 = F$.

At $\sqrt{8t} = \ell$, define

$$\mathcal{F}_\ell(A) = \ell^{-4} \int_B |G_t|^2 d^4x, \quad K_\eta = \{A : \mathcal{F}_\ell(A) \geq \eta^2/\ell^4\}.$$

The constraint is gauge invariant. It measures a flowed block L^2 norm. It does **not** give a supremum or an adjoint-curvature operator norm bound. In particular, it cannot be substituted for the curvature upper bound used in the Jacobian estimates of the Jacobian note or the corrected instability note, §1 (relevant passages read). The original phrase “up to replacing a supremum by a block average” concealed an additional estimate; that identification is withdrawn.

2. The bound and the conditions for equality

Proposition 1 (continuum action cost). Under the assumptions of Section 1, for every $\ell > 0$ and $\eta \geq 0$,

$$s(A) \geq \frac{\eta^2}{4g^2} \quad (A \in K_\eta).$$

The infimum has the same lower bound; if the set is empty its infimum is understood as $+\infty$.

Proof. Use the ungauged flow

$$\partial_s(B_s)_\nu = D_\mu(G_s)_{\mu\nu} =: J_{s,\nu}.$$

Variation of $s = (4g^2)^{-1} \int \sum_{a,\mu,\nu} (G_{\mu\nu}^a)^2$ and integration by parts give

$$\delta s = -\frac{1}{g^2} \int \sum_\nu \langle J_{s,\nu}, \delta B_\nu \rangle d^4x, \quad \frac{d}{ds} s(B_s) = -\frac{1}{g^2} \|J_s\|_2^2 \leq 0.$$

The norm in the last display sums the colour and one-form indices. The factor $1/g^2$ is explicit for this flow-time convention; the old formula $dS_E/ds = -\|D^*G\|_2^2$ omitted the action prefactor. The continuum flow convention is the one recorded in Lüscher’s local companion (introductory flow equations, passage read); the displayed variation checks the normalization directly.

Consequently

$$s(A) \geq s(B_t) = \frac{1}{4g^2} \int_\Omega |G_t|^2 \geq \frac{1}{4g^2} \int_B |G_t|^2 \geq \frac{\eta^2}{4g^2}.$$

The second inequality is positivity outside B ; the last is the constraint. $\square$

Scope of uniformity. The same algebraic bound applies at each admissible ℓ and domain volume; it does not assert flow existence or regularity uniformly through any limit. It uses a fixed invariant inner product and action coefficient. Writing the action in this form for different compact gauge groups gives the same coefficient by **normalization convention**, not a comparison theorem for their measures, mode counts or operators. Rescaling the inner product also rescales the field norm and the threshold. The former claim for every lattice spacing $a < \ell$ is withdrawn: an exact Wilson-lattice analogue requires a specified lattice observable, its action comparison and its flow monotonicity, none of which is proved by this continuum integral.

Proposition 2 (equality accounting; replacement for withdrawn sharpness). For every $A \in K_\eta$ in Proposition 1,

$$\begin{aligned} s(A) - \frac{\eta^2}{4g^2} &= \frac{1}{g^2} \int_0^t \|J_s\|_2^2 ds + \frac{1}{4g^2} \int_{\Omega \setminus B} |G_t|^2 d^4x \\ &\quad + \frac{1}{4g^2} \left(\int_B |G_t|^2 d^4x - \eta^2 \right). \end{aligned}$$

Each term is nonnegative. Equality therefore holds if and only if all three vanish: zero flow dissipation, zero flowed curvature outside B , and exact saturation of the block threshold.

Proof. Integrate the dissipation identity, split the final action integral into B and its complement, and subtract the threshold. Nonnegativity gives the equivalence. For a smooth flow, zero dissipation entails $J_s = 0$ throughout the interval. $\square$

This is an identity and necessary-and-sufficient equality test, **not** an existence assertion. For $\eta = 0$ a flat connection gives equality when admitted by the domain. For $\eta > 0$, neither an

equality example nor a sequence approaching equality is constructed here, so the optimality of $1/4$ remains unproved. Strict inequality for particular fields would not itself give a uniform improvement of the infimum.

Why the old example fails. Prescribing a nonzero constant abelian curvature on B and zero outside does not produce a smooth admissible connection. In that sector the Bianchi identity is $dF = 0$, whereas $d(\mathbf{1}_B F_{\text{const}})$ has nonzero face-supported terms for a nonzero constant two-form on a bounded cube. A smooth cutoff of the connection can give an admissible localized field, but then its curvature includes transition terms whose cost and flow must be controlled. No such control is supplied by the former example.

Nor does a/ℓ control flow at the radius of the block. Already in the abelian heat-flow model, rescaling $x = \ell y$ leaves the smoothing time $t/\ell^2 = 1/8$ fixed. The heat kernel samples a boundary layer of width comparable with ℓ , so a relative effect of order one is possible as $a/\ell \rightarrow 0$. There is no vanishing cutoff correction justified by that limit. These observations invalidate the old sharpness proof without claiming that no other sharpness proof exists.

3. Instability and the block constraint are separate questions

The corrected instability note, §§2–3 (full note read) derives, in a stationary abelian chromomagnetic background with charged curvature $b = gB > 0$, the linear growth rate

$$b - k_{\parallel}^2 \quad (k_{\parallel}^2 < b).$$

The full covariant Laplacian is included. The maximal model rate is b , not $2b$. This is a background spectral calculation, not an admissible block-supported saturation example for Proposition 1.

Correction to the descent argument. Monotonicity says $s(\Phi_t A) \leq s(A)$ for each initial connection A ; it does not compare the flowed block functionals of two different initial connections. From $s(A') < s(A)$ one cannot conclude $\mathcal{F}_{\ell}(A') < \mathcal{F}_{\ell}(A)$. The action can also change outside the block and during the flow. Even decreasing $\mathcal{F}_{\ell}$ would not force exit from K_{η} for an interior point with threshold slack. The claims that every descent direction exits K_{η} and that a constant field minimizes on it are withdrawn.

Proposition 1 holds for every admissible field in the constraint set, regardless of its stability. It supplies a floor on the constrained action, not a minimizer or its Hessian. Neither homogeneity nor inhomogeneity of a constrained minimizer is established here. No Gaussian expansion about such a minimizer is justified by this note.

4. What the ingredients do and do not estimate

quantity	corrected value or status	support and limitation
continuum action cost	$s(A) \geq \eta^2/(4g^2)$	Proposition 1; sharpness open
unstable-mode phase-space count	model $b^2 \ell^4/(8\pi^2)$	Constant charged-background Landau density times longitudinal momentum-disc density; not an exact finite-block count or general entropy bound
linearized-flow growth	model $e^{(b-k_{\parallel}^2)t}$; majorant $e^{2t\Gamma}$	Corrected instability note; $\Gamma = \frac{1}{2} \sup_{s,x} \ M(s,x)\ _{\text{op}}$, $Mu = 2[G,u]$; no sharp truncation-tail claim

quantity	corrected value or status	support and limitation
normalized large-field weight	not proved	Needs a constrained/full partition-function comparison; a mode count alone supplies none

The count in the entropy note, §2 (passage read) multiplies the densities $b/(2\pi)$ and $b/(4\pi)$ by the transverse and longitudinal areas. It is a **phase-space model** for one charged branch with its stated multiplicity convention. Finite-domain boundary conditions, flux compatibility, discrete momenta and colour multiplicities matter. The parameter $b\ell^2$ is not automatically the threshold η of our colour and ordered-index L^2 norm. Thus neither an exact count $\eta^2/(8\pi^2)$ nor a general cost/count ratio $2\pi^2/g^2$ is proved here. A convergent polymer sum requires additional estimates beyond this model.

Likewise Γ is an upper bound on a different norm, not the block threshold. If separately $\Gamma \leq \eta_\infty/\ell^2$, then at $t = \ell^2/8$ the corrected kernel growth majorant is $e^{2t\Gamma} \leq e^{\eta_\infty/4}$. This is conditional on the heat-equation hypotheses of the corrected instability note; it is not an error lower bound or a Hamiltonian truncation theorem. No such upper bound follows from membership in K_η .

Missing normalization. In a finite regulator with reference probability measure ν , write $d\mathbb{P} = Z^{-1}e^{-s}d\nu$. If an action bound $s \geq q$ holds on an event K , then only

$$\mathbb{P}(K) \leq Z^{-1}e^{-q}\nu(K)$$

follows immediately. This elementary inequality does not establish a uniform local tail: the partition-function denominator still needs control. This is the correction in ground-state measure transfer, §3 (passage read). The continuum action calculation itself defines no quantum probability measure.

The former net exponent

$$-\frac{\eta^2}{4g^2} \left(1 - \frac{Cg^2}{2\pi^2}\right)$$

is retained only as **heuristic bookkeeping**, obtained by multiplying an assumed suppression $e^{-\eta^2/(4g^2)}$ by an assumed entropy factor $e^{C\eta^2/(8\pi^2)}$. Neither factor as a normalized interacting measure estimate, nor a uniform C , is established here. Negativity of that expression would not by itself prove summability over blocks or polymers. The earlier statement that the first three rows were proved or sharp and that the large-field corner was closed is withdrawn.

Finally, a measure estimate, even if subsequently proved, does not supply a signed Hamiltonian operator comparison. That distinction is explained in the operator-inequality note, §3 (passage read); its additional localization estimates are not used here. This note proves no spectral or physical mass gap.

5. Consequence for STATE

Retain the continuum action floor $\eta^2/(4g^2)$ for the flowed block L^2 constraint, subject to the stated flow and boundary assumptions. Sharpness, constrained minimizers, an exact lattice analogue, normalized probability control and large-field polymer convergence remain unproved here. The constant-background mode count and the net exponent are model/heuristic inputs only; the sharp $e^{2\eta}$ claim is withdrawn in line with the later instability correction. The large-field corner remains an estimate and construction problem, not an estimate closure. No continuum $SU(3)$ construction or mass gap is obtained. This correction specifies the scope for the caller's STATE and catalog updates; those files are unchanged in this task.